A Synopsis

of

Project Based Learning IV

CCE2270

Pattern-Based NLP for Mental Health Analysis

Submitted to Manipal University Jaipur

towards the partial fulfillment of the requirement for the award of the degree of

BACHELOR OF TECHNOLOGY

in

Computer and Communication Engineering

Jan – May 2026

By:

Laraib Firoz (23FE10CCE00052)

Pragati Khapra (23FE10CCE00069)

Upasana Dwivedy (23FE10CCE00127)

MANIPAL UNIVERSITY JAIPUR

Under the guidance of:

Amit Sharma

Department of Computer and Communication Engineering

Manipal University Jaipur

INTRODUCTION

Mental health has become a critical area of concern in modern society, especially with the rapid growth of digital communication platforms where individuals increasingly express their thoughts, emotions, and daily experiences through text. Natural Language Processing (NLP) has emerged as a powerful tool for analysing such textual data, enabling automated understanding of emotional states, sentiment, and psychological cues from written language. As a result, NLP-based systems are being widely explored for mental health monitoring and emotional well-being support.

However, most existing NLP approaches for mental health analysis rely on single-message or snapshot based analysis, where each text input is evaluated independently. These systems typically classify individual messages into predefined emotional or sentiment categories and generate immediate responses. While this method is useful for detecting short-term emotions, it does not align well with the nature of human mental health, which is rarely defined by isolated moments. Psychological distress often develops gradually and manifests through repeated language patterns, persistent negative expressions, reduced emotional diversity, and difficulty recovering from negative emotional states over time.

Focusing solely on isolated messages can lead to incomplete or misleading interpretations of an individual's mental state. For example, a single neutral or positive message may mask an ongoing pattern of distress, resulting in false reassurance. This limitation highlights the need for more advanced analytical approaches that consider language as a time evolving signal rather than a collection of independent text samples.

This project addresses these challenges by proposing a pattern-based NLP framework for mental health analysis. Instead of evaluating messages in isolation, the system examines linguistic and emotional patterns across multiple interactions over time. By tracking trends such as emotional persistence, repetition of negative thoughts, and changes in expressive richness, the proposed approach aims to capture sustained psychological distress more accurately.

The importance of this topic lies in its potential to improve the reliability, explainability, and safety of mental health-related AI systems. By emphasizing longitudinal patterns rather than momentary emotions, the project contributes toward more meaningful mental health monitoring that supports early awareness and informed interventions without attempting clinical diagnosis.

MOTIVATION

The motivation for this research arises from the growing reliance on text-based communication and the increasing use of artificial intelligence in mental health support systems. People frequently express their emotional states, stress, and personal struggles through digital text such as messages, online posts, and personal journals. This makes language a valuable source for understanding mental well-being. However, despite advances in NLP, many existing mental health analysis systems still rely on short-term, message-level predictions that do not reflect how psychological distress develops and persists.

Mental health conditions such as chronic stress, anxiety, and depression are not defined by isolated emotional expressions but by patterns that emerge over time. Repeated negative thoughts, persistent pessimism, emotional stagnation, and slow recovery from negative emotions are key indicators of sustained distress. Systems that analyse single messages often fail to detect these patterns, leading to unreliable outputs, false reassurance, or oversimplified interpretations of complex mental states.

This project is important because it shifts the focus from snapshot-based emotion detection to longitudinal pattern analysis, which better aligns with real psychological processes. By modelling emotional and linguistic trends across multiple interactions, the proposed approach can identify early warning signals of prolonged distress rather than reacting to momentary mood changes. This makes the system more meaningful for long-term mental well-being monitoring.

The potential impact of this work lies in its ability to produce explainable and interpretable insights instead of black-box predictions. Rather than assigning labels or diagnoses, the system summarizes observed patterns such as persistent negativity or increasing repetition in language, making the results safer and easier to understand. This approach can support wellness applications, self-monitoring tools, and research-oriented mental health systems where early awareness and trend understanding are more valuable than instant classification.

Overall, the project is motivated by the need for more reliable, psychologically grounded, and responsible AI systems that respect the complexity of mental health and provide insights based on sustained behavioural patterns rather than isolated textual signals.

PROBLEM STATEMENT

Problem Definition

Existing NLP-based mental health analysis systems primarily rely on snapshot-based techniques that analyse individual text messages in isolation. These systems typically perform sentiment or emotion classification on single inputs and generate immediate predictions. While such methods can identify short-term emotional states, they fail to capture the longitudinal psychological patterns that reflect sustained mental distress. As a result, important indicators such as emotional persistence, repetitive negative thinking, and gradual emotional decline remain undetected.

Mental health conditions develop progressively and are reflected through repeated linguistic behaviours rather than isolated expressions. Treating each message as an independent data point leads to incomplete, and sometimes misleading, interpretations of an individual's mental state. This creates a gap between how mental health manifests in real life and how current NLP systems model it.

Objectives of the Project

The primary objectives of this project are:

- To design a pattern-based NLP framework that analyses text data across multiple interactions over time.
- To identify longitudinal linguistic and emotional patterns such as repeated negative expressions, emotional persistence, and reduced expressive variation.
- To move beyond single-message sentiment classification and model language as a time-evolving signal.
- To generate interpretable summaries that describe observed emotional trends instead of instant labels or diagnoses.
- To improve the reliability and psychological relevance of AI-driven mental health analysis systems.

Pros of Existing Methods:

- Simple and computationally efficient to implement.
- Effective for detecting immediate emotions or short-term mood changes.
- Suitable for real-time response systems and basic emotional feedback.

Cons of Existing Methods:

- Analyse messages independently without considering historical context.
- Fail to detect sustained psychological distress or gradual emotional changes.
- Prone to false reassurance due to isolated positive or neutral messages.
- Provide limited explainability and oversimplified emotional interpretations.

Problem Addressed

This research addresses the limitations of existing snapshot-based NLP systems by introducing a longitudinal, pattern-based approach. By focusing on emotional and linguistic trends over time, the proposed system aims to bridge the gap between real-world mental health behaviour and AI-based text analysis, enabling safer, more accurate, and more meaningful mental health monitoring.

LITERATURE REVIEW

Depression is one of the most common mental health disorders worldwide, and early detection plays a crucial role in effective treatment. Traditional diagnosis methods depend on self-reported questionnaires and clinical interviews, which are time-consuming and subjective. With the advancement of Artificial Intelligence, researchers have started using Natural Language Processing (NLP) techniques to automatically detect depression from spoken or written language. One of the most widely used datasets for this purpose is the DAIC-WOZ (Distress Analysis Interview Corpus – Wizard of Oz) dataset.

The DAIC-WOZ dataset contains interview conversations between participants and a virtual interviewer designed to simulate a clinical setting. Along with interview transcripts, the dataset provides depression severity scores such as PHQ-8 and PHQ-9. Due to the availability of clean transcripts and standard labels, DAIC-WOZ has become a benchmark dataset for depression detection research.

Early studies using DAIC-WOZ focused on simple text-based features such as word frequency, sentence length, and use of negative or emotional words. These features were combined with traditional machine learning algorithms like Support Vector Machines and Logistic Regression. Results showed that depressed individuals often use more negative language, self-focused words, and shorter responses. Although these methods provided initial success, they required manual feature design and had limited accuracy.

With the rise of deep learning, researchers started using neural network models such as LSTM and CNN to analyze interview transcripts. These models were able to capture patterns in language over longer conversations and showed improved performance compared to traditional methods. However, due to the small size of the DAIC-WOZ dataset, deep learning models often suffered from overfitting and lack of robustness.

More recent research applies transformer-based models such as BERT to DAIC-WOZ transcripts. These models use pre-trained language knowledge and can better understand context and meaning in sentences. Transformer-based approaches have shown higher accuracy in detecting depression and predicting severity scores compared to earlier techniques. Some studies also combine text with audio and facial features to further improve performance.

Despite promising results, several limitations remain in existing work. Most studies rely only on questionnaire scores as ground truth, which may not fully represent clinical depression. Many models are tested only on the DAIC-WOZ dataset and may not generalize well to real-world conversations. In addition, most approaches act as black boxes and do not clearly explain why a person is classified as depressed.

Research Gap

Although significant progress has been made in detecting depression using NLP techniques on datasets such as DAIC-WOZ, several important research gaps remain. Most existing approaches prioritize improving prediction accuracy through complex machine learning and deep learning models, often treating the task as a pure classification problem. While these models achieve high performance metrics, they frequently operate as black-box systems, offering limited insight into how linguistic features relate to underlying depressive symptoms.

There is a lack of focused research on identifying and interpreting specific language patterns that correspond to different manifestations of depression, such as persistent negative thinking, reduced emotional expression, or repetitive pessimistic language. Without this interpretability, it becomes difficult to assess the clinical relevance and trustworthiness of model outputs, especially in sensitive mental health contexts where explanations are as important as predictions.

Additionally, most existing studies rely on controlled interview-style datasets, including DAIC-WOZ, where participants respond to structured prompts. Models trained in such environments may not generalize effectively to real-world scenarios such as informal conversations, personal journals, or chat-based interactions. This raises concerns about the robustness and applicability of current systems outside laboratory conditions.

Furthermore, limited attention has been given to longitudinal language analysis, where changes and patterns across multiple interactions are examined instead of isolated responses. This gap restricts the ability of NLP systems to capture sustained depressive tendencies that develop gradually over time.

Therefore, there is a clear need for NLP-based depression detection approaches that are simple, interpretable, and robust, focusing on pattern-based language analysis rather than only accuracy-driven classification. Addressing these gaps can lead to more reliable and clinically meaningful mental health analysis systems that are better suited for real-world applications.

METHODOLOGY

This research adopts a systematic methodology to investigate pattern-based language analysis for depression detection using NLP, addressing the limitations identified in existing literature. The focus is not only on prediction accuracy but also on interpretability, longitudinal analysis, and real-world relevance.

1. Dataset Selection and Understanding: The DAIC-WOZ (Distress Analysis Interview Corpus – Wizard of Oz) dataset is used as the primary data source for this research. The dataset consists of interview transcripts between participants and a virtual interviewer, along with depression severity labels such as PHQ-8 and PHQ-9 scores. DAIC-WOZ is selected due to its widespread use in depression detection research and availability of standardized annotations, enabling fair comparison with existing approaches.

2. Data Preprocessing: Text transcripts are pre-processed to ensure consistency and reliability. This includes removal of irrelevant symbols, normalization of text, tokenization, and handling of missing or incomplete responses. Speaker turns are preserved to maintain conversational structure, which is important for analysing language patterns across interactions.

3. Baseline Per-Message Analysis: As a reference point, per-message sentiment and emotion analysis is performed using pretrained NLP models. Basic linguistic features such as sentence length, word frequency, use of self-referential terms, and presence of negative emotional words are extracted. This step reflects the snapshot-based methods commonly used in existing literature and serves as a baseline for comparison.

4. Pattern-Based Feature Extraction: To address the research gap, the study shifts from isolated message analysis to longitudinal pattern analysis. Linguistic and emotional features are aggregated across multiple responses from each participant. Temporal patterns such as persistent negative sentiment, repetition of pessimistic expressions, reduced vocabulary diversity, and slow recovery from negative language are identified. These features are designed to reflect sustained depressive tendencies rather than momentary emotional states.

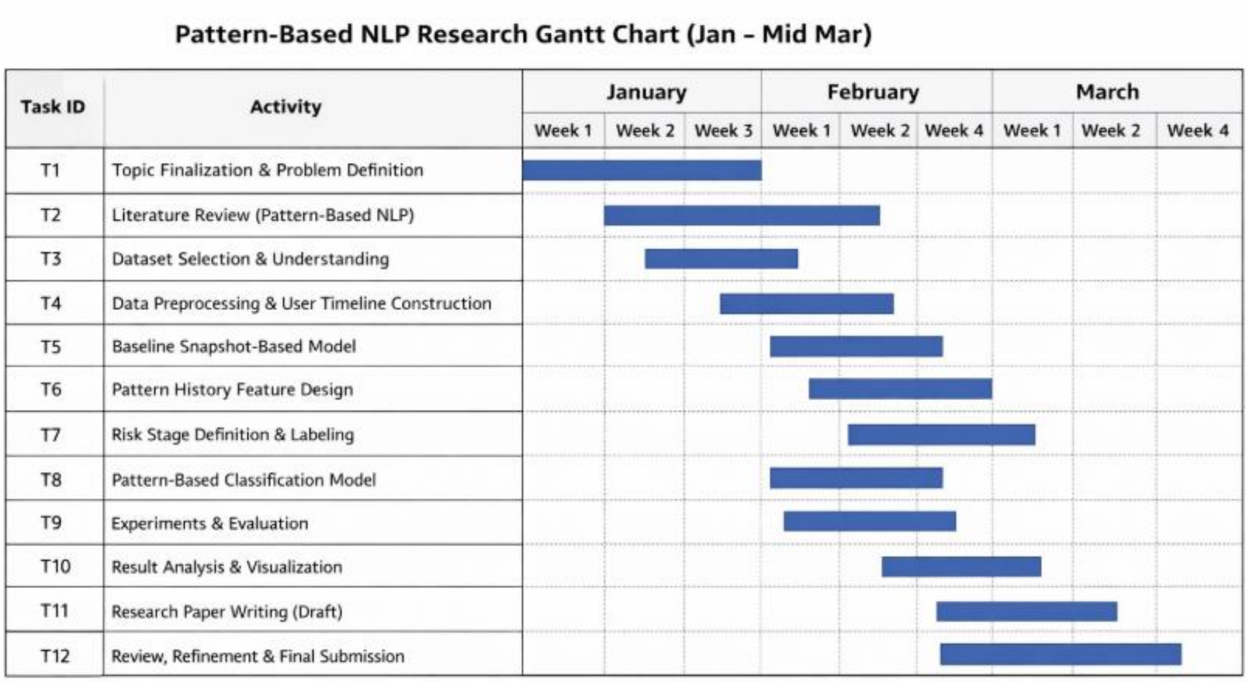
5. Model Design and Analysis: Instead of relying solely on complex black-box models, simpler and interpretable modelling techniques are explored to analyse the extracted patterns. The focus is on understanding which language patterns contribute most to depression detection and how they relate to known depressive symptoms, rather than maximizing raw accuracy alone.

6. Evaluation Strategy: The proposed pattern-based approach is evaluated by comparing its performance with traditional snapshot-based methods. Evaluation criteria include prediction performance, consistency across time, robustness to variation in responses, and interpretability of results. PHQ-based labels are used for quantitative comparison while acknowledging their limitations as clinical proxies.

7. Interpretation and Result Discussion: The results are analysed to determine whether longitudinal language patterns provide clearer and more reliable indicators of depression compared to single-message analysis. Emphasis is placed on explainable outputs, such as descriptive summaries of observed language behaviour, rather than diagnostic conclusions.

8. Documentation and Conclusion: Finally, findings are documented with a discussion of how pattern-based NLP addresses the limitations identified in prior research. The study concludes by highlighting its contributions, limitations, and potential directions for future work, including extension to real-world conversational data beyond structured interview settings.

Gantt Chart



FACILITIES REQUIRED

The successful execution of the proposed research requires standard computational facilities and commonly used software tools suitable for Natural Language Processing and machine learning experiments. No specialized or clinical-grade equipment is required, as the study is based on secondary textual data analysis.

Software Requirements

- **Operating System:** Windows / Linux / macOS
- **Programming Language:** Python
- **Development Environment:** Jupyter Notebook or Visual Studio Code
- **NLP Libraries:** NLTK, spaCy, Hugging Face Transformers
- **Machine Learning Libraries:** Scikit-learn, TensorFlow or PyTorch
- **Data Processing Tools:** NumPy, Pandas
- **Visualization Tools:** Matplotlib, Seaborn
- **Dataset Access:** DAIC-WOZ dataset for interview transcripts and depression labels

Hardware Requirements

- **Processor:** Intel i5 / AMD Ryzen 5 or equivalent
- **RAM:** Minimum 8 GB (16 GB recommended for model training)
- **Storage:** At least 10–20 GB free disk space for datasets and models
- **GPU (Optional):** NVIDIA GPU with CUDA support for faster training of deep learning models

Other Requirements

- **Internet Connectivity:** Required for accessing datasets, pretrained language models, and research resources
- **Documentation Tools:** MS Word / LaTeX for report preparation

Initial Work Done

At the current stage, the work carried out for this project is limited to an extensive literature review. A detailed study of existing research related to depression detection using Natural Language Processing (NLP) was conducted, with particular focus on studies using the DAIC-WOZ dataset. Various approaches including traditional machine learning methods, deep learning models, transformer-based architectures, and multimodal techniques were reviewed and compared.

Through this review, key linguistic patterns associated with depression, commonly used preprocessing techniques, model architectures, evaluation metrics, and existing limitations in current research were identified. The study also helped in understanding the strengths and weaknesses of different methodologies and highlighted important research gaps such as lack of interpretability, limited generalization, and over-reliance on questionnaire-based labels.

No model implementation, prototype development, or experimental simulation has been performed at this stage. However, the literature review has provided a strong theoretical foundation and clear direction for the proposed methodology. Based on the insights gained, the problem statement and objectives of the project have been refined.

Challenges Faced and Solutions

The main challenge encountered was the wide variation in methodologies and evaluation protocols across existing studies, making direct comparison difficult. This was addressed by categorizing the reviewed works based on model type, input features, and evaluation strategy, allowing a structured understanding of the research landscape.

EXPECTED OUTCOME

The proposed research is expected to demonstrate that pattern-based, longitudinal analysis of language provides more meaningful insights into depression detection than traditional snapshot-based NLP approaches. By analysing linguistic and emotional trends across multiple interactions, the system is anticipated to capture sustained depressive tendencies that are often missed by single-message classification methods.

The project is expected to identify specific language patterns associated with depressive symptoms, such as persistent negative expressions, repetitive pessimistic phrasing, reduced emotional and vocabulary variation, and slow recovery from negative language. These findings will help establish clearer links between linguistic behaviour and underlying depressive tendencies, improving interpretability and clinical relevance.

Compared to existing black-box models, the proposed approach is expected to produce more explainable outputs in the form of descriptive summaries of observed language patterns rather than opaque predictions. This can enhance trust, transparency, and usability in mental health–related NLP applications.

The research is also anticipated to show that simple and interpretable pattern-based features can achieve competitive performance while being more robust and better aligned with real-world mental health scenarios. By focusing on longitudinal language behaviour, the project contributes toward bridging the gap between controlled interview datasets such as DAIC-WOZ and practical mental health monitoring applications.

Overall, the expected outcome of this work is a structured and interpretable NLP framework that advances depression detection research by emphasizing pattern-based analysis, explainability, and long-term emotional trends, thereby contributing to safer and more clinically meaningful AI-driven mental health systems.

BIBLIOGRAPHY / REFERENCES

1. Gratch, J., Artstein, R., Lucas, G., Stratou, G., Scherer, S., Nazarian, A., Wood, R., Boberg, J., DeVault, D., Marsella, S., Traum, D., Rizzo, A., & Morency, L.-P. (2014).
   The Distress Analysis Interview Corpus of human and computer interviews.
   Proceedings of LREC 2014.

2. DeVault, D., Artstein, R., Benn, G., Dey, T., Fast, E., Gainer, A., Georgila, K., Gratch, J., Hartholt, A., Lhommet, M., Lucas, G., Marsella, S., Morency, L.-P., Nazarian, A., Scherer, S., Stratou, G., Suri, A., Traum, D., Wood, R., & Xu, Y. (2014).
   SimSensei Kiosk: A virtual human interviewer for healthcare decision support.
   Proceedings of AAMAS.

3. American Psychiatric Association. (2013).
   Diagnostic and Statistical Manual of Mental Disorders (DSM-5).
   APA Publishing.

4. Kroenke, K., Spitzer, R. L., & Williams, J. B. (2001).
   The PHQ-9: Validity of a brief depression severity measure.
   Journal of General Internal Medicine.

5. Kroenke, K., & Spitzer, R. L. (2002).
   The PHQ-9: A new depression diagnostic and severity measure.
   Psychiatric Annals.

6. Pennebaker, J. W., Boyd, R. L., Jordan, K., & Blackburn, K. (2015).
   The development and psychometric properties of LIWC2015.
   University of Texas at Austin.

7. Morales, M. R., Scherer, S., Gratch, J., & Morency, L.-P. (2018).
   Towards multimodal automated depression analysis.
   IEEE Transactions on Affective Computing.

8. Cummins, N., Scherer, S., Krajewski, J., Schnieder, S., Epps, J., & Quatieri, T. (2015).
   A review of depression and suicide risk assessment using speech analysis.
   Speech Communication.

9. Ringeval, F., Schuller, B., Valstar, M., Cowie, R., Kaya, H., & Pantic, M. (2017).
   AVEC 2017 – Real-life depression, emotion and affect recognition workshop.
   Proceedings of ACM MM.

10. Trotzek, M., Koitka, S., & Friedrich, C. M. (2018).
    Utilizing neural networks and linguistic metadata for early detection of depression.
    Proceedings of ACL.

11. Huang, K., Guntuku, S. C., Narayanan, S., & Choudhury, M. D. (2020).
    Understanding the language of mental health on social media.
    ACL Workshop on Computational Linguistics and Clinical Psychology.

12. Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2019).
    BERT: Pre-training of deep bidirectional transformers for language understanding.
    NAACL-HLT.

13. Liu, Y., Ott, M., Goyal, N., Du, J., Joshi, M., Chen, D., Levy, O., Lewis, M., Zettlemoyer, L., & Stoyanov, V. (2019).
    RoBERTa: A robustly optimized BERT pretraining approach.
    arXiv preprint arXiv:1907.11692.

14. Alhanai, T., Ghassemi, M., & Glass, J. (2018).
    Detecting depression with audio/text sequence modeling of interviews.
    Interspeech.

15. Low, D. M., Bentley, K. H., & Ghosh, S. S. (2020).
    Automated assessment of psychiatric disorders using speech: A systematic review.
    Laryngoscope Investigative Otolaryngology.

16. Haque, A., Guo, M., Miner, A., & Fei-Fei, L. (2018).
Measuring depression symptom severity from spoken language and facial expressions.
arXiv preprint arXiv:1811.08592.

17. Sadeghi, M., Schuller, B., & Morency, L.-P. (2024).
Harnessing multimodal large language models for depression severity assessment.
Scientific Reports.

18. Valstar, M., Schuller, B., Smith, K., Eyben, F., Jiang, B., Bilakhia, S., & Pantic, M. (2013).
AVEC 2013: The continuous audio/visual emotion and depression recognition challenge.
ACM Multimedia.

19. Chancellor, S., & De Choudhury, M. (2020).
Methods in predictive techniques for mental health status on social media.
ACM Computing Surveys.

20. Shatte, A. B. R., Hutchinson, D. M., & Teague, S. J. (2019).
Machine learning in mental health: A scoping review.
Psychological Medicine.